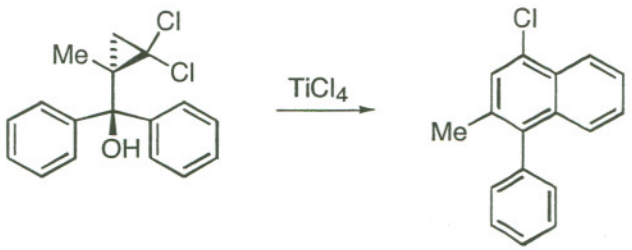
Organic Cume
April 16, 2004

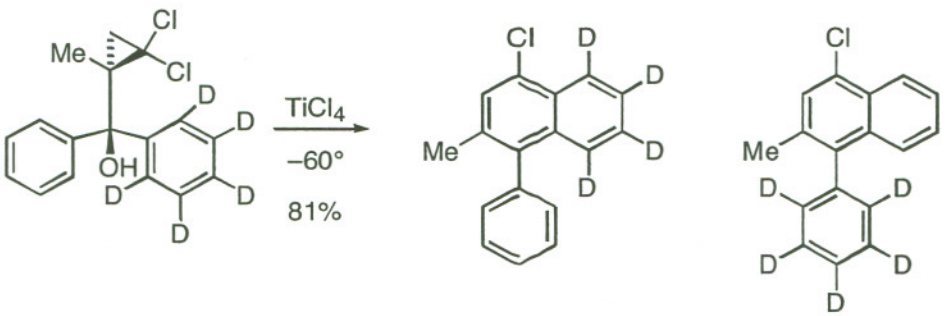
Use of molecular models on this cumulative exam is permitted, and may be helpful.

A well known and much studied rearrangement of cyclopropylmethanols is the solvolysis shown below, with fragmentation of the ring to make a butenyl cation:

1. The following transformation begins with just such a fragmentation. Provide a mechanism.



2. Now, amplify your mechanism with stereochemical arguments to explain the fact that the following compound gives mostly the product having deuterium in the naphthalene ring. Hint: this is not an isotope effect; instead, think of $TiCl_4$ as a chelating Lewis acid.



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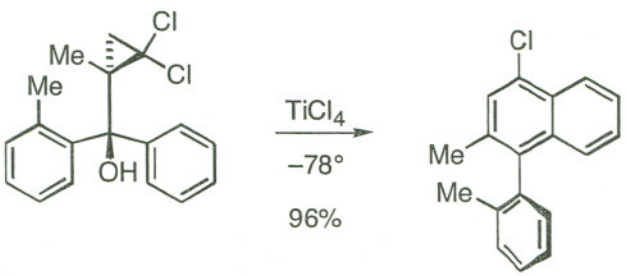
3. In a final twist, explain the chirality transfer from central to axial chirality in the following example.



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